

Keebox KBox Entity Specification

Definitions of KBox, KCredentialBox, and KNoteBox

1. Purpose

This document defines the three core data concepts used by Keebox: KBox, KCredentialBox, and KNoteBox. It describes what each concept represents, how the concepts relate to one another, and the role each plays in the product. It intentionally does not define implementation code, database schemas, cryptography, synchronization, or transport details.

2. KBox

KBox is the common conceptual base for secure user-owned content in Keebox. It represents a protected logical container whose data belongs to one Keebox user and is managed by the application as private information.

KBox is not intended to describe one specific kind of secret. Instead, it establishes the shared identity and behavior expected from all protected Keebox box types. Specialized box types extend this concept according to the kind of information they contain.

Core characteristics of a KBox

- It belongs to a specific Keebox user.
- It represents private content that must remain protected.
- It is primarily stored on the user's device.
- It can participate in Keebox backup and synchronization.
- It has a stable identity so that the same logical box can be tracked across local and remote storage.
- It acts as the parent concept for specialized Keebox box types.

The current specialized KBox types are KCredentialBox and KNoteBox.

3. KCredentialBox

KCredentialBox is the KBox type dedicated to credentials and secret authentication-related information. It represents a secure unit of credential data owned by a user.

A KCredentialBox can represent information associated with an account, service, application, system, development environment, infrastructure resource, or any other context where secret credentials must be retained securely.

Typical information represented by a KCredentialBox

- Email addresses and account usernames.
- Passwords and passphrases.
- API keys and API secrets.

- Access tokens and secret tokens.
- Security keys or other application-specific secret values.
- Service or account labels needed to identify the credential.
- Additional private credential-related text maintained by the user.

A KCredentialBox is not limited to a fixed provider such as an email platform or cloud service. It is a general Keebox concept for securely representing user credentials.

4. KNoteBox

KNoteBox is the KBox type dedicated to private text notes. It represents text that the user wants Keebox to retain as confidential information rather than as a credential.

A KNoteBox may contain short or long-form private text. The meaning of the text is defined by the user; Keebox treats the content as private data that must be protected and available locally.

Typical uses of a KNoteBox

- Private personal notes.
- Recovery information or confidential reminders.
- Sensitive project notes.
- Private technical information.
- Any text that the user does not want stored as ordinary unprotected notes.

5. Relationship Between the Types

KCredentialBox and KNoteBox are specialized forms of KBox. They share the common concept of user ownership, privacy, protected storage, stable identity, and participation in backup or synchronization. They differ in the semantic meaning of the data they represent.

Concept	Purpose	Examples
KBox	Base concept for protected Keebox user data.	Parent concept for all specialized KBox types.
KCredentialBox	Represents credentials and authentication secrets.	Passwords, API keys, tokens, account secrets.
KNoteBox	Represents private text notes.	Confidential notes, recovery text, sensitive reminders.

6. Conceptual Hierarchy

The current Keebox data hierarchy is:

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KBox
| - KCredentialBox
| - KNoteBox
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Additional specialized KBox types can be introduced in the future without changing the meaning of the existing types, provided they preserve the common KBox principles of ownership, privacy, protected storage, and consistent synchronization identity.